

DEVESH GIRI

 DeveshGiri |  deveshgiri |  deveshgiri.github.io |  devesh.giri@students.iiserpune.ac.in

EDUCATION

2021-2026 BS-MS at **Indian Institute of Science Education and Research, Pune, IN**
2021 Class 12th (AISSCE) at **Disha Delphi Public School, Kota, Rajasthan**
2019 Class 10th (ICSE) at **H. P. Children's Academy, Gorakhpur, U.P.**

PUBLICATIONS ([INSPIREHEP](#)/[ARXIV](#)/[GOOGLE SCHOLAR](#))

Giri, Devesh and Suvodip Mukherjee (2026). “How lonely are the Binary Compact Objects Detected by the LIGO-Virgo-KAGRA Collaboration?” In: *arXiv:2604.22441*.

Giri, Devesh and Bhooshan Gadre (2025). “Sub-Solar Mass Intermediate Mass Ratio Inspirals: Waveform Systematics and Detection Prospects with Gravitational Waves”. In: *arXiv:2511.13324*.

Giri, Devesh, Sajal Mukherjee, et al. (2026). “The effects of eccentricity on GWs from IMRIs in the deci-hertz band”. In: *prep*.

PROJECTS/SOFTWARE/CODE RELEASES

playgwtc

[Link to Read the Docs documentation](#)

It is an open-source user-friendly Python command-line tool for fetching, processing, and visualizing data for gravitational-wave events from the Gravitational Wave Open Science Center. This tool allows you to browse GW events through prefix-search, instantly generate high-quality plots, including time-frequency Q-transforms of the raw detector strain data from chosen detectors, and theoretical waveform models based on the event's physical parameters. Installable through pip: [Link to PyPI page](#)

EXPERIENCE

Tata Institute of Fundamental Research, Mumbai, IN

June 2025 - March 2026

Visiting Master's Thesis Student

Guide: [Dr. Suvodip Mukherjee](#)

- Modeled effects of three-body encounters on gravitational radiation from black hole binaries.
- Quantified ‘dephasing’ of *perturbed* waveforms with respect to the vacuum waveforms and noted differences in observable quantities.
- Constraining the three-body perturber parameter space for the events in the Gravitational Wave Transient Catalog (GWTC) 4.0.

IUCAA, Pune, IN and BITS-Pilani, IN

June 2025 - present

Research Internship

Guides: [Dr. Sajal Mukherjee](#), [Dr. Apratim Ganguly](#)

- Studied the FastEMRIWaveforms (FEW) formalism for IMRIs/EMRIs.
- Quantified the role of eccentricity for IMRI systems in the deci-hertz GW frequency band using mismatch and fitting-factor calculations.
- Studied the excitation of different radial modes under equatorial eccentric orbits for IMRI systems.

Universitat Politècnica de Valencia (UPV), Valencia

February 2025 - May 2025

Research Internship

Guide: [Prof. Pau Amaro-Seoane](#)

- On formation channels for IMBH GW sources and cluster complex landscape.

- Derived analytic and semi-analytic relations for estimating the timescale for the first merger in a set of many stellar clusters, with a fraction of them hosting IMBHs each.
- Used approaches in dynamical interactions, encounter rates and combinatorics to compare and match the results against numerical gravitational N-body simulations.

Nikhef, Amsterdam, NL

October 2023 - December 2024

Research Internship

Guide: Dr. Bhooshan Gadre

- The paper can be found here: [arXiv:2511.13324](https://arxiv.org/abs/2511.13324) (currently under review in *Physical Review D*)
- Quantified the detectability and waveform systematics aspects of sub-solar mass intermediate mass ratio inspirals (SSM-IMRIs) with the current and future GW detectors.
- Performed in-depth fitting factor studies using differential evolution and Bayesian parameter estimation studies to understand parameter biases/error, and waveform systematics. Familiarized with IMRPHENOMX and BHPTNRSURROGATE waveform models.
- Demonstrated the significant effects of higher harmonics for IMRI systems and need to develop aligned and precessing spin waveform surrogates.

Indian Institute of Technology, Bombay, IN

May 2024 – July 2024

Krittika Summer Project (KSP 5.0)

Mentors: Krittika Eclipsing Binaries Group

- Co-authored a comprehensive collaborative report ([link to the report](#)) with eight contributors, covering the classical two-body problem; the physics of binary star systems with emphasis on spectroscopic and eclipsing binaries; forward modeling and inverse parameter estimation for eclipsing binaries; and their analysis using PHOEBE (PHysics Of Eclipsing BinariEs).

Vanderbilt University, Nashville, USA

May 2023 - July 2023

Summer Internship

Guide: Dr. Karan Jani

- Completed PyCBC tutorials and familiarized with using PyCBC Inference and Bilby for bayesian parameter estimation.
- Worked on studying the improvement in parameter estimates by including LIGO-India in the detector network.
- Demonstrated 30% – 90% improvements in different parameters through bayesian parameter estimation for GW190521-like systems.

SCHOOLS/LECTURES/COURSES

CIERA, Northwestern University, IL, USA (online)

August 2025

Code/Astro 2025 A Software Engineering Workshop for Astronomy
Sarah Blunt

Guides: Dr. Jason Wang, Dr.

- Hands-on tutorials and lectures on Development Environments, Programming Paradigms, Git Mechanics, Debugging, Parallel Programming, `jax`, Releasing Code, Code Documentation, Software Testing, Anti-Discriminatory Practices, and Code Profiling.
- As a part of the workshop project, released an open-source (pip installable) software `playgwtc` with command-line tools for fetching, processing, and visualizing data for gravitational-wave events from the Gravitational Wave Open Science Center.

International Centre for Theoretical Sciences - TIFR, Bengaluru, IN

July 2025

GW Summer School *Guides: Dr. Surhud More, Dr. Chiara Caprini, Dr. Tirthankar Roy Choudhury*

- Undertook three courses: Probing the early universe using GW observations; Cosmography using GW standard sirens; and, Probing large-scale structure using GW observations
- Besides the rigorous theoretical lectures, gained practical experience in GW Population Analysis, Cosmological Parameter Inference and N-Body Cosmological Simulations. Hands-on experience with different softwares/codes used for such studies: GWCOSMO, ICAROGW and GADGET-4
- Delivered a flash talk and presented a poster on SSM-IMRIs.

- An overview of multi-wavelength astronomy (with qualitative descriptions of astrophysical objects), mechanism of radio telescopes and GMRT.
- Learned to work with FITS files and about mean and median stacking with some optimising techniques like the Binapprox algorithm.
- An exposure of equatorial/Galactic coordinates.
- An exposure of Astronomy Catalogues. Worked with AT20G BSS catalogue and SuperCOSMOS all-sky catalogue to learn about cross-matching.
- Learned about optimisation and used NumPy optimisation, Sorting optimisation and Binary search optimisation.
- Learned about k-d trees optimisation using Astropy

The University of Sydney, Australia (Coursera)

- Course conducted by Dr. Tara Murphy and Dr. Simon Murphy, focussing on working with large datasets, implementing algorithms, and learning from data using ML tools. I learnt to work with files from exoplanet surveys, pulsar detections, and galaxy clusters, using Python and SQL.

SKILLS

Programming languages and softwares	Python, Bash, Git, L ^A T _E X, MATLAB, PyCBC, BILBY, GWCOSMO, ICAROGW, GADGET-4
Operating systems	macOS, Linux, Windows
Cluster computing	HTCondor

CONFERENCES/TALKS/SEMINARS

National Science Day, TIFR-Mumbai	February'26, 28
DAIM-DAA, TIFR-Mumbai	January'26, 19 - January'26, 20
Science Fest, TIFR-Mumbai	December'25, 19
The Future of Gravitational-Wave Astronomy, ICTS-TIFR	October'25, 27 - October'25, 31
Pune-Mumbai Cosmology & Astro-Particle meeting	September'25, 12 - September'25, 13
33 rd IAGRG Conference, BITS-Pilani	January'25, 02 - January'25, 04

WORKSHOPS

Workshop on Gravitational Waves and LIGO-India <i>BITS, Pilani</i>	October 2024
Gravitational Wave Open Data Workshop gwosc.org and <i>IUCAA</i>	April 2024
Workshop on Data Science in Astronomy <i>IUCAA</i>	December 2023

VOLUNTEER EXPERIENCE

Presenter

November 22, 2025

Frontiers of Science, 2025

TIFR, Mumbai, IN

- Presented the research areas in GW Astrophysics on the behalf of < data|theory > Universe Lab in the annual one day program at TIFR's Colaba campus. Running for more than a decade, the program comprises of lab visits and demonstrations curated to give glimpses into modern research.

Core Member and Tutor

August 2022 - April 2023

Mindspark, DISHA

IISER, Pune, IN

- Mindspark, DISHA is an initiative by the students of IISER-Pune which imparts education to underprivileged students of Grade 8 by conducting classes every Saturday and Sunday. I worked to design the structure of Mathematics lectures and also volunteered for weekly lectures as a tutor.

REFeree DETAILS

Dr. Suvodip Mukherjee

Assistant Professor (Reader),
Department of Astronomy and Astrophysics,
Tata Institute of Fundamental Research (TIFR), Mumbai, IN
Email: suvodip@tifr.res.in
Website: <https://web.tifr.res.in/suvodip.mukherjee/>

Dr. Bhooshan Gadre

Assistant Professor (Scientist E),
Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, IN
Email: bhooshan.gadre@iucaa.in
Website: <https://www.iucaa.in/en/faculty-research/Bhooshan-Gadre>

Dr. Sajal Mukherjee

Assistant Professor,
Department of Physics,
Birla Institute of Technology & Science Pilani (BITS-Pilani), Pilani, IN
Email: sajal.mukherjee@pilani.bits-pilani.ac.in
Website: <https://www.bits-pilani.ac.in/pilani/sajalmukherjee/>

Dr. Apratim Ganguly

Research Faculty,
Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune, IN
Email: apratim@iucaa.in
Website: <https://inspirehep.net/authors/1298756>